



Advanced Time Series Analysis and Forecasting

Seminar 10: Long memory and rough volatility



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ **ACF**: Autocorrelation Function
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model); in event studies: Abnormal Return
- ▣ **ARCH**: AutoRegressive Conditional Heteroskedasticity
- ▣ **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ **ARMA**: AutoRegressive Moving Average (model)
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **BUX**: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ **CAC**: Cotation Assistée en Continu (CAC 40 index, Paris) — the CAC 40 index of the Paris bourse
- ▣ **CEE**: Central and Eastern Europe
- ▣ **DM**: Diebold–Mariano (test)
- ▣ **ELW**: Exact Local Whittle (estimator)
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **FCVAR**: Fractionally Cointegrated VAR
- ▣ **FFT**: Fast Fourier Transform
- ▣ **FIGARCH**: Fractionally Integrated GARCH

Acronyms used in this material (2/2)

- ▣ **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GPH**: Geweke–Porter–Hudak (estimator)
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HAR**: Heterogeneous AutoRegressive (model)
- ▣ **LW**: Local Whittle (estimator)
- ▣ **MSE**: Mean Squared Error
- ▣ **OLS**: Ordinary Least Squares
- ▣ **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ **QLIKE**: Quasi-LIKElihood loss
- ▣ **RFSV**: Rough Fractional Stochastic Volatility (model)
- ▣ **RV**: Realised Variance
- ▣ **SD**: Standard Deviation
- ▣ **SE**: Standard Error
- ▣ **VECM**: Vector Error Correction Model
- ▣ **WIG20**: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's question and route

- **Question:** how do we measure, test and use long memory and roughness in volatility and inflation?
 - ▶ this seminar comes **before** Lecture 10: the section "Prerequisites for Today" gives every definition the tasks use
- **Route**
 - ▶ Part A: ARFIMA autocorrelations, fractional weights, standard errors and bias of the estimators, fractional Gaussian noise, the RFSV kernel, circulant embedding, switching that mimics long memory
 - ▶ Part B: memory of FTSE 100 realised variance, BET and EUR/RON; roughness of the Euro Stoxx 50 and of Bitcoin; forecasting FTSE 100 and CAC 40 realised variance
 - ▶ Part C: co-memory of DAX and CAC 40 volatility; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	ARFIMA autocorrelations and the mean; fractional weights	Solved, Proposed	A1
A3, A4	standard errors of GPH and LW; bias and bandwidth	Solved, Proposed	A3
A5, A6	fractional Gaussian noise; the RFSV kernel	Solved, Proposed	A5
A7, A8	circulant embedding by hand; switching that mimics long memory	Solved, Proposed	A7, A1
B1, B2	memory of FTSE 100 log RV; BET and EUR/RON with the Qu test	Solved, Proposed	B1
B3, B4	roughness of the Euro Stoxx 50; Bitcoin by sampling interval	Solved, Proposed	B3
B5, B6	forecasting RV of the FTSE 100 and the CAC 40	Solved, Proposed	B5
C1, C2	fractional cointegration of two volatilities; what is wrong in an AI answer?	Proposed	B1, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
RV of FTSE 100, Euro Stoxx 50, DAX, CAC 40	Heber et al. (2009)	daily, 5-minute returns	2000–2022
RV of Bitcoin	Binance one-minute prices	daily, 1–60 minutes	2018–2026
BET	EODHD	daily	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026

- ▣ RV in $\%^2$, returns in %
- ▣ Functions in the notebook
 - ▶ `gph`, `local_whittle`, `elw`, `qu_stat`, `gjr_scaling`, `h_with_noise`, `rfsv_kernel`, `forecast_rv`, `fcvar_fit`
- ▣ No account or key is needed

Prerequisites for Today (1/24): autocorrelation and memory

- **Autocovariance** and **autocorrelation** of a stationary series X_t at lag k

$$\gamma(k) = \text{Cov}(X_t, X_{t+k}), \quad \rho(k) = \frac{\gamma(k)}{\gamma(0)} \in [-1, 1]$$

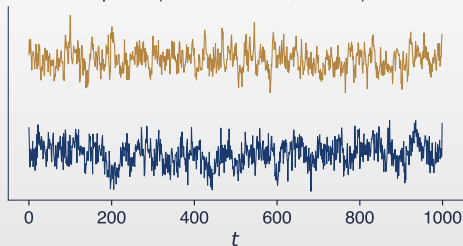
- ▶ $\gamma(0) = \text{Var}X_t$; stationary: mean and $\gamma(k)$ do not depend on t
- **Short memory**: autocorrelations decay exponentially and are summable
 - ▶ AR(1) $X_t = \phi X_{t-1} + \varepsilon_t$, $|\phi| < 1$: $\rho(k) = \phi^k$, so $\sum_k |\rho(k)| < \infty$
 - ▶ ε_t : white noise (uncorrelated, mean 0, variance σ^2)
- **Long memory**: hyperbolic decay, not summable, a spectral pole at frequency zero

$$\rho(k) \sim c k^{2d-1} \ (k \rightarrow \infty), \quad f(\lambda) \sim G \lambda^{-2d} \ (\lambda \rightarrow 0), \quad 0 < d < 1/2$$

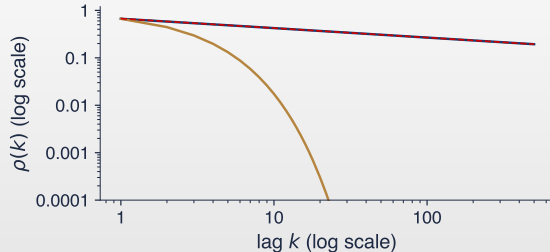
- ▶ d : memory parameter; $c, G > 0$: constants; $f(\lambda)$: spectral density at frequency λ (defined below)
- ▶ $a_k \sim b_k$: the ratio $a_k/b_k \rightarrow 1$; Hurst exponent $H = d + 1/2$

Prerequisites for Today (2/24): hyperbolic against exponential decay

Simulated paths (same variance, shifted)



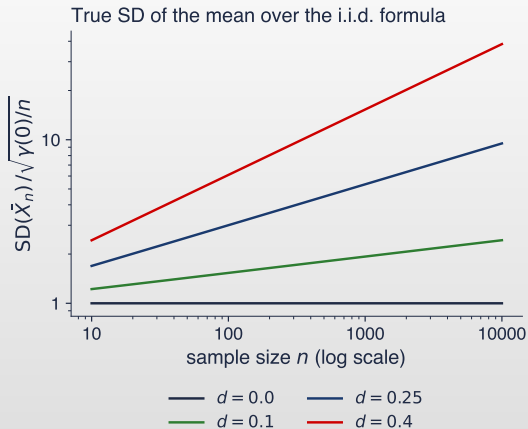
Autocorrelations



— AR(1), $\phi = 0.67$ — ARFIMA(0, 0.4, 0) - - - $\frac{\Gamma(1-d)}{\Gamma(d)} k^{2d-1}$

- ARFIMA(0, 0.4, 0) (defined below) and an AR(1) with the same $\rho(1)$, $\phi = 0.67$; both simulated, $n = 1000$
- At lag 100: $\rho = 0.27$ for ARFIMA, practically 0 for the AR(1); on log scales the hyperbolic ACF is a straight line of slope $2d - 1$
- The ARFIMA path wanders in long swings around its mean

Prerequisites for Today (3/24): the variance of the sample mean



- For any stationary series

$$\text{Var}(\bar{X}_n) = \frac{1}{n} \sum_{|k| < n} \left(1 - \frac{|k|}{n}\right) \gamma(k)$$

- ▶ $\bar{X}_n = \frac{1}{n} \sum_{t=1}^n X_t$; n : sample size
- ▶ i.i.d.: only $k = 0$ remains, $\gamma(0)/n$
- Long memory: of order n^{2d-1} , not n^{-1}
 - ▶ the ratio in the chart grows like n^d
 - ▶ $n = 1000$: 1.9, 5.3, 15.3 for $d = 0.1, 0.25, 0.4$
 - ▶ the i.i.d. standard error is that many times too small

Prerequisites for Today (4/24): the lag operator and fractional differencing

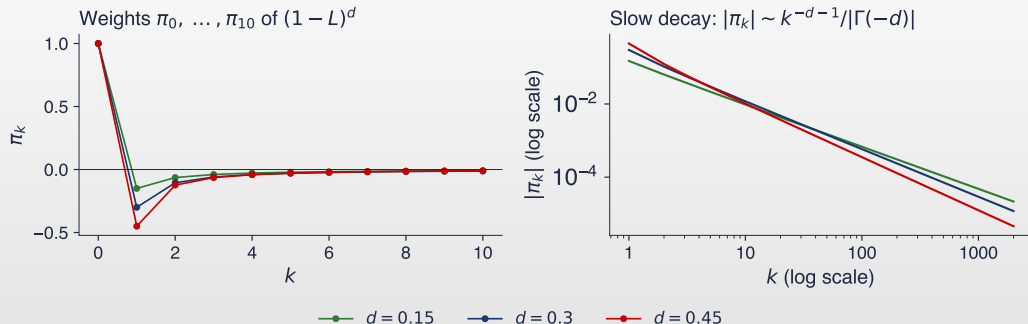
□ **Lag operator** L : $LX_t = X_{t-1}$, $L^k X_t = X_{t-k}$; first difference $(1 - L)X_t = X_t - X_{t-1}$

□ **Fractional difference**: the binomial series of $(1 - L)^d$ for a real d

$$(1 - L)^d X_t = \sum_{k=0}^{\infty} \pi_k X_{t-k}, \quad \pi_0 = 1, \quad \pi_k = \pi_{k-1} \frac{k-1-d}{k}$$

- ▶ π_k : weight of X_{t-k} ; $d = 1$: $\pi_1 = -1$, $\pi_k = 0$ for $k \geq 2$ (the usual difference)
- ▶ $0 < d < 1$: all past values enter, with weights $\pi_k \sim k^{-d-1}/\Gamma(-d)$
- ▶ $\Gamma(z) = \int_0^{\infty} u^{z-1} e^{-u} du$: gamma function, $\Gamma(n) = (n-1)!$; $\Gamma(-d) < 0$ for $0 < d < 1$
- ▶ $\sum_k \pi_k = (1-1)^d = 0$: a constant level is removed only by the infinite filter

Prerequisites for Today (5/24): fractional weights



- Left: $\pi_1 = -d$, then small negative weights that shrink slowly
- Right: on log scales $|\pi_k|$ is a straight line of slope $-(d+1)$: a hyperbolic, not exponential, decay
- A filter truncated after K lags leaves out the tail of the weights

Prerequisites for Today (6/24): the ARFIMA model

- **ARFIMA**(0, d , 0) [Hosking \(1981\)](#): white noise after fractional differencing

$$(1 - L)^d X_t = \varepsilon_t, \quad -\frac{1}{2} < d < \frac{1}{2} \text{ (stationary and invertible)}$$

- ▶ $d > 0$: long memory; $d < 0$: antipersistence ($\sum_k \rho(k) = 0$); $d \geq 1/2$: nonstationary

- Autocorrelations: a recursion and its limit

$$\rho(0) = 1, \quad \rho(k) = \rho(k-1) \frac{k-1+d}{k-d}, \quad \rho(k) \sim \frac{\Gamma(1-d)}{\Gamma(d)} k^{2d-1}$$

- **ARFIMA**(1, d , 0): $(1 - \phi L)(1 - L)^d X_t = \varepsilon_t$

- ▶ ϕ : AR coefficient, $|\phi| < 1$; it changes the short lags, d the long ones
- ▶ **ARFIMA**(p , d , q): an **ARMA**(p , q) for $(1 - L)^d X_t$

Prerequisites for Today (7/24): the spectral density and the periodogram

- **Spectral density:** the variance of X_t decomposed by frequency

$$f(\lambda) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \gamma(k) e^{-i\lambda k}, \quad \gamma(0) = \int_{-\pi}^{\pi} f(\lambda) d\lambda$$

- ▶ $\lambda \in [-\pi, \pi]$: angular frequency, a cycle of $2\pi/\lambda$ periods; low λ : long cycles
- ▶ ARFIMA(0, d , 0): $f(\lambda) = \frac{\sigma^2}{2\pi} (4 \sin^2(\lambda/2))^{-d} \sim \frac{\sigma^2}{2\pi} \lambda^{-2d}$

- **Periodogram** at the Fourier frequencies: the sample counterpart of f

$$I(\lambda_j) = \frac{1}{2\pi n} \left| \sum_{t=1}^n X_t e^{-i\lambda_j t} \right|^2, \quad \lambda_j = \frac{2\pi j}{n}, \quad j = 1, \dots, \lfloor n/2 \rfloor$$

- ▶ $I(\lambda_j)/f(\lambda_j)$ is approximately exponential with mean 1, independent across j : very noisy
- ▶ bandwidth m : the number of lowest frequencies $\lambda_1, \dots, \lambda_m$ used to estimate d ; $m \rightarrow \infty$, $m/n \rightarrow 0$

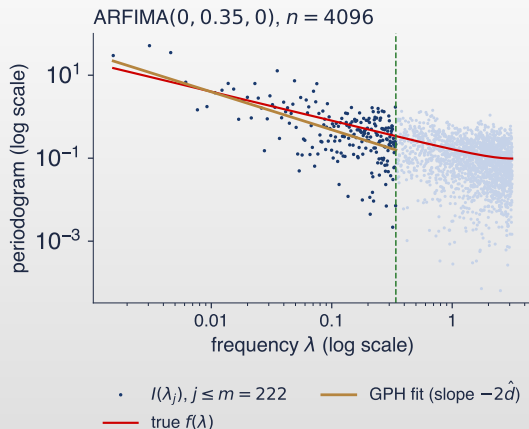
Prerequisites for Today (8/24): the GPH estimator

- **GPH** Geweke and Porter-Hudak (1983): take logs of $I(\lambda_j) \approx f(\lambda_j) \xi_j$ near zero and run OLS

$$\log I(\lambda_j) = a + d \left[-\log(4 \sin^2(\lambda_j/2)) \right] + u_j, \quad j = 1, \dots, m$$

- ▶ ξ_j : exponential error with mean 1; $u_j = \log \xi_j - \mathbb{E} \log \xi_j$; a : intercept
- ▶ $\hat{d}$: the OLS slope; regressor $\approx -2 \log \lambda_j$, so the log-log slope is $-2d$
- Asymptotic standard error
 - ▶ $\text{Var} u_j = \pi^2/6$ and the regressor has variance ≈ 4 : $\text{SE}(\hat{d}) = \pi/\sqrt{24m}$
 - ▶ $\sqrt{m}(\hat{d} - d) \rightarrow N(0, \pi^2/24)$; 95% interval $\hat{d} \pm 1.96 \text{ SE}$
- Precision grows only with $\sqrt{m}$, not with $\sqrt{n}$: estimating d needs long samples

Prerequisites for Today (9/24): the log-periodogram near zero



- Simulated ARFIMA(0, 0.35, 0), $n = 4096$
- The $m = n^{0.65} = 222$ lowest ordinates (dark) lie around a line
 - ▶ GPH: $\hat{d} = 0.46$; local Whittle: 0.44
 - ▶ both within two SEs of $d = 0.35$
- The scatter around the line is the exponential noise ξ_j : it does not shrink with n
- Far from zero (pale points) the short-run dynamics bend f : they must stay out of the regression

Prerequisites for Today (10/24): the local Whittle estimator

- **Local Whittle Robinson (1995b)**: Gaussian likelihood of $f(\lambda_j) = G\lambda_j^{-2d}$ on the m lowest frequencies

$$\hat{d} = \arg \min_d R(d), \quad R(d) = \log \hat{G}(d) - \frac{2d}{m} \sum_{j=1}^m \log \lambda_j, \quad \hat{G}(d) = \frac{1}{m} \sum_{j=1}^m \lambda_j^{2d} I(\lambda_j)$$

- ▶ $\hat{G}(d)$: the level G that fits best for a given d ; $R(d)$: the profiled objective (a QLIKE-type fit)

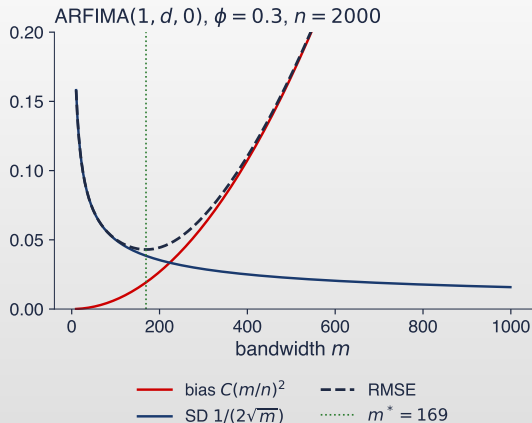
- Inference

- ▶ $\sqrt{m}(\hat{d} - d) \rightarrow N(0, 1/4)$: $SE = 1/(2\sqrt{m})$, smaller than GPH at the same m
- ▶ relative efficiency: the ratio of the two asymptotic variances, $(\pi^2/24)/(1/4)$
- ▶ consistent for $-1/2 < d \leq 1$; for $d > 1$ it converges to 1

- **Exact local Whittle (ELW) Shimotsu and Phillips (2005), Shimotsu (2010)**

- ▶ replaces $\lambda_j^{2d} I(\lambda_j)$ by the periodogram of $(1 - L)^d(X_t - \hat{\mu})$, $\hat{\mu}$: estimated mean
- ▶ $N(d, 1/(4m))$ for any d in the search range, also $d \geq 1/2$ (nonstationary)

Prerequisites for Today (11/24): bias and the choice of bandwidth



- Short-run dynamics bias $\hat{d}$; for ARFIMA(1, d , 0)

$$\mathbb{E}\hat{d} - d \approx C\left(\frac{m}{n}\right)^2, \quad C = \frac{2\pi^2}{9} \cdot \frac{2\phi}{(1-\phi)^2}$$

- $\phi > 0$: positive bias, growing with m

- MSE = bias² + $\frac{1}{4m}$; minimised at

$$m^* = \left(\frac{n^4}{16C^2}\right)^{1/5}$$

- MSE: mean squared error; m^* grows like $n^{4/5}$
- chart: $\phi = 0.3$, $n = 2000$: $C = 2.69$,
 $m^* = 169 = n^{0.68}$

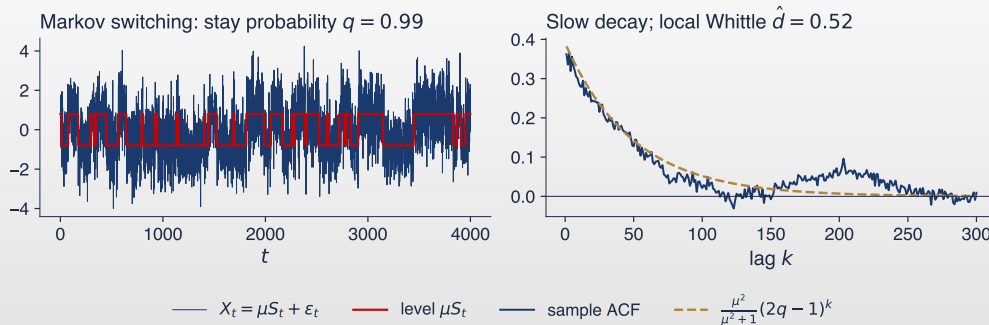
Prerequisites for Today (12/24): level shifts that mimic long memory

- **Two-state Markov chain** $S_t \in \{-1, 1\}$: it stays with probability q , switches with $1 - q$

$$X_t = \mu S_t + \varepsilon_t, \quad \mathbb{E}(S_t S_{t+k}) = (2q - 1)^k$$

- ▶ μ : size of the level; ε_t : i.i.d. $N(0, 1)$, independent of S ; expected duration of a regime $1/(1 - q)$
- ▶ the ACF decays exponentially: short memory in the strict sense
- **Half-life** of a geometric decay $\rho(k) = \rho(0) r^k$
 - ▶ $k_{1/2} = \ln 0.5 / \ln r$: the lag at which the autocorrelation has halved
- Rare shifts, structural breaks and slowly varying trends put power at the lowest frequencies [Diebold and Inoue \(2001\)](#)
 - ▶ $\hat{d}$ is then large although the process has short memory: **spurious long memory**

Prerequisites for Today (13/24): a simulated switching series



- $q = 0.99$ (regimes last 100 periods on average), $\mu = 0.8$, $n = 4000$
- Local Whittle with $m = n^{0.65}$ gives $\hat{d} = 0.52$, a value typical of a long-memory series
- Over a few hundred lags the slow geometric decay cannot be told apart from a hyperbolic one

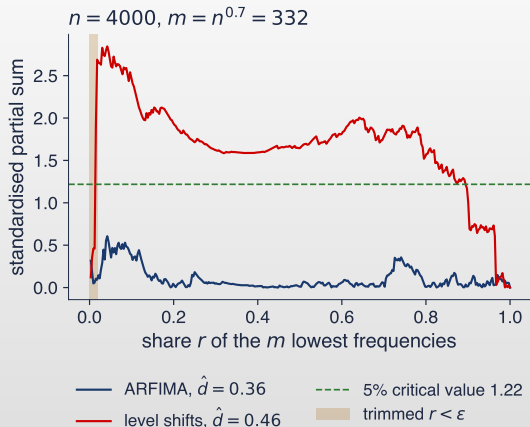
Prerequisites for Today (14/24): the Qu test

- **Qu test** [Qu \(2011\)](#): H_0 stationary long memory, $f(\lambda) = G\lambda^{-2d}$ on the m lowest frequencies

$$W = \sup_{r \in [\varepsilon, 1]} \left(\sum_{j=1}^m \nu_j^2 \right)^{-1/2} \left| \sum_{j \leq mr} \nu_j \left(\frac{\lambda_j^{2\hat{d}} I(\lambda_j)}{\hat{G}} - 1 \right) \right|$$

- ▶ $\hat{d}$, $\hat{G} = \hat{G}(\hat{d})$: local Whittle estimates; $\nu_j = \log \lambda_j - \frac{1}{m} \sum_i \log \lambda_i$
 - ▶ the bracket is the score of local Whittle at frequency j ; its full sum is zero at $\hat{d}$
 - ▶ r : share of the m frequencies summed; ε : trimming, the first εm partial sums are skipped
- Reading
- ▶ under H_0 the partial sums stay small at every r ; level shifts make the lowest ordinates too large: a big W
 - ▶ reject for W above the critical value; with $m = n^{0.7}$, $\varepsilon = 0.02$ (simulated): 1.09 (10%), 1.22 (5%), 1.56 (1%)

Prerequisites for Today (15/24): the Qu partial sums



- Two simulated series, $n = 4000$
- ARFIMA(0, 0.35, 0): the curve stays low
 - ▶ $W = 0.61$: do not reject
- Markov level shifts ($q = 0.997, \mu = 0.8$)
 - ▶ $\hat{d}$ looks like long memory
 - ▶ but the curve jumps at small r : $W = 2.85$, reject at 1%
- A rejection says only that stationary long memory does not fit: breaks, shifts or trends are alternatives

Prerequisites for Today (16/24): fractional cointegration

- **Fractional cointegration:** p series, each with memory d , whose combinations $\beta'X_t$ have memory $d - b$, $b > 0$
- **FCVAR Johansen and Nielsen (2012):** a VECM written in fractional differences

$$\Delta^d X_t = \alpha \beta' L_b \Delta^{d-b} X_t + \sum_{i=1}^k \Gamma_i \Delta^d L_b^i X_t + \varepsilon_t, \quad \Delta^d = (1 - L)^d, \quad L_b = 1 - \Delta^b$$

- ▶ β ($p \times r$): cointegrating vectors; α ($p \times r$): adjustment speeds; r : cointegration rank
- ▶ L_b : fractional lag operator (only past values); k : number of lagged short-run terms, Γ_i their matrices; $b = d = 1$: the usual VECM
- Estimation and rank tests
 - ▶ for fixed (d, b) : Johansen reduced-rank regression; profile likelihood: maximise over a grid of (d, b)
 - ▶ trace test of rank r against p : $LR = 2(\ell_p - \ell_r)$, ℓ : maximised log likelihoods; $\chi^2_{(p-r)^2}$ when $b < 1/2$

Prerequisites for Today (17/24): fractional Brownian motion

- **Fractional Brownian motion** (fBm) W_t^H [Mandelbrot and Van Ness \(1968\)](#): Gaussian, $W_0^H = 0$, mean 0, covariance

$$\mathbb{E}(W_t^H W_s^H) = \frac{1}{2} (t^{2H} + s^{2H} - |t - s|^{2H}), \quad \mathbb{E}|W_{t+\Delta}^H - W_t^H|^2 = \Delta^{2H}$$

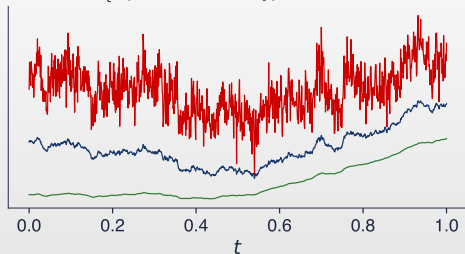
- ▶ $H \in (0, 1)$: Hurst exponent; $H = 1/2$: Brownian motion, independent increments
- ▶ $H < 1/2$: rough (irregular) paths, negatively correlated increments; $H > 1/2$: smooth paths, persistent increments

- **Fractional Gaussian noise** (fGn): the unit increments $W_{t+1}^H - W_t^H$

$$\gamma_k = \rho(k) = \frac{1}{2} (|k+1|^{2H} - 2|k|^{2H} + |k-1|^{2H})$$

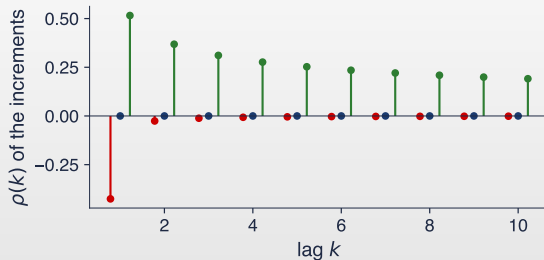
- ▶ $\rho(1) = 2^{2H-1} - 1$; for $H > 1/2$, $\rho(k) \sim H(2H-1)k^{2H-2}$: long memory with $d = H - 1/2$

Prerequisites for Today (18/24): rough and smooth paths

Paths of W_t^H (shifted vertically)

— $H = 0.1$ — $H = 0.5$ — $H = 0.8$

Fractional Gaussian noise



- Left: fBm paths from the same random numbers for $H = 0.1, 0.5, 0.8$
- Right: $H = 0.1$ gives $\rho(1) \approx -0.43$ and almost nothing beyond: the path reverts at once
- $H = 0.8$: positive, slowly decaying autocorrelations, the long memory of the previous frames

Prerequisites for Today (19/24): circulant embedding

- **Goal:** simulate n points of a stationary Gaussian series with autocovariances $\gamma_0, \dots, \gamma_{n-1}$ exactly
 - ▶ Cholesky factorisation of the $n \times n$ covariance costs $O(n^3)$
- **Davies–Harte** [Davies and Harte \(1987\)](#): embed the covariance in a circulant matrix of size $M = 2(n - 1)$

$$c = (\gamma_0, \gamma_1, \dots, \gamma_{n-1}, \gamma_{n-2}, \dots, \gamma_1), \quad \lambda_k = \sum_{j=0}^{M-1} c_j e^{-2\pi i j k / M}$$

- ▶ circulant: each row is the previous one shifted by one place; c : its first row
- ▶ λ_k : its eigenvalues, the FFT (fast Fourier transform) of c , real because c is symmetric
- ▶ if all $\lambda_k \geq 0$: $Y = \text{FFT}(\sqrt{\lambda/M} Z)$, Z complex Gaussian; the first n values have the target covariance
- ▶ cost $O(n \log n)$; for fGn all $\lambda_k \geq 0$ for every H

Prerequisites for Today (20/24): realised variance and moment scaling

- ▣ **Realised variance** of day t from N_t intraday returns $r_{t,i}$ (e.g. 5-minute)

$$RV_t = \sum_{i=1}^{N_t} r_{t,i}^2, \quad \sigma_t = \sqrt{RV_t}$$

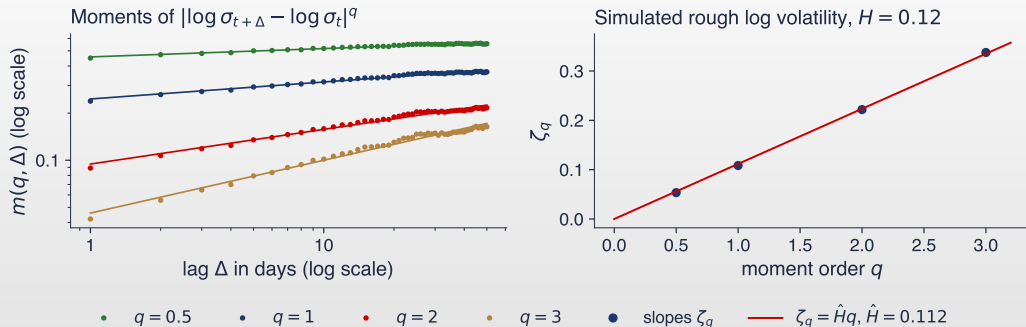
- ▶ a noisy measure of the daily variance; fewer returns per day: a larger measurement error

- ▣ **Moment scaling** Gatheral, Jaisson and Rosenbaum (2018): for $\log \sigma_t = \nu W_t^H$ the moments of the increments are power laws

$$m(q, \Delta) = \frac{1}{N} \sum_t |\log \sigma_{t+\Delta} - \log \sigma_t|^q \propto \Delta^{\zeta_q}, \quad \zeta_q = qH$$

- ▶ q : moment order; Δ : lag in days; N : number of pairs; ν : volatility of volatility (vol-of-vol)
- ▶ ζ_q : slope of $\log m(q, \Delta)$ on $\log \Delta$; then $\hat{H} = \sum_q q \zeta_q / \sum_q q^2$ (regression through the origin)
- ▶ i.i.d. measurement error of variance s^2 adds an intercept: $m(2, \Delta) = \nu^2 \Delta^{2H} + 2s^2$

Prerequisites for Today (21/24): the scaling regression



- Simulated daily log volatility $0.3 W_t^H$ with $H = 0.12$, 4000 days, no measurement error
- Left: straight lines on log scales, one per q ; right: the slopes lie on a line through the origin, $\hat{H} = 0.112$
- Departures from linearity in q (multifractality, heavy-tailed increments) show up at large q

Prerequisites for Today (22/24): HAR, FIGARCH and lognormal forecasts

- **HAR Corsi (2009)**: daily, weekly and monthly averages of the past, $y_t = \log RV_t$

$$y_{t+h} = \beta_0 + \beta_d y_t + \beta_w \bar{y}_t^{(5)} + \beta_m \bar{y}_t^{(22)} + u_{t+h}, \quad \bar{y}_t^{(K)} = \frac{1}{K} \sum_{j=0}^{K-1} y_{t-j}$$

- ▶ h : horizon; OLS; a short-memory AR(22) with restricted weights that mimics long memory

- **FIGARCH Baillie, Bollerslev and Mikkelsen (1996)**: the GARCH variance with a fractional difference of the squared shocks e_t^2

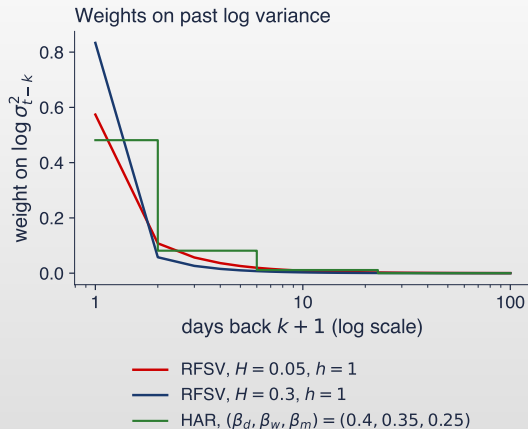
$$\sigma_t^2 = \omega + [1 - (1 - \beta L)^{-1} (1 - \varphi L)(1 - L)^d] e_t^2$$

- ▶ $\omega > 0$, β , φ : GARCH-type coefficients; the bracket is an ARCH(∞) filter whose weights sum to one

- **Lognormal correction**: models forecast $\log RV$, the target is RV

- ▶ if $\log RV_{t+h} \sim N(\mu, s^2)$: $\mathbb{E}RV_{t+h} = e^{\mu+s^2/2}$; e^μ alone is the median, too low

Prerequisites for Today (23/24): the RFSV predictor



- RFSV: $\log \sigma_t = \nu W_t^H$; the forecast is a power-kernel average of the past

$$\widehat{\log \sigma_{t+h}^2} = \sum_k w_k \log \sigma_{t-k}^2$$

- ▶ $w_k \propto \int_k^{k+1} \frac{du}{(u+h)u^{H+1/2}}$, normalised to sum to 1; h : horizon

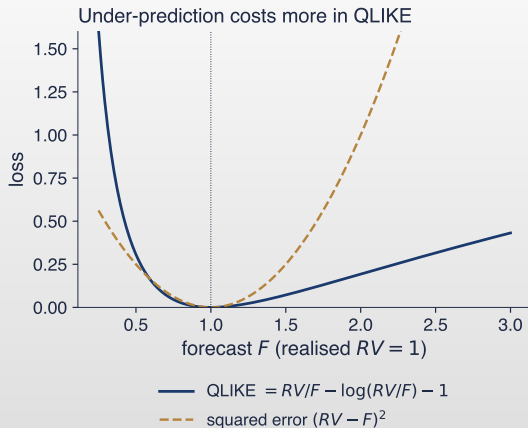
- Variance forecast

$$\exp(\widehat{\log \sigma_{t+h}^2} + 2c\nu^2 h^{2H})$$

- ▶ $c = \frac{\Gamma(3/2-H)}{\Gamma(H+1/2)\Gamma(2-2H)}$; $4c\nu^2 h^{2H}$: the forecast-error variance of $\log \sigma^2$, halved as in the lognormal correction

- Chart: the kernel puts most weight on the last day and a long thin tail behind; HAR uses three flat steps

Prerequisites for Today (24/24): QLIKE and the Diebold–Mariano test



- **QLIKE** [Patton \(2011\)](#) of a variance forecast F

$$\text{QLIKE} = \frac{RV}{F} - \log \frac{RV}{F} - 1 \geq 0$$

- ▶ 0 only if $F = RV$; ranks forecasts correctly although RV is a noisy proxy
- ▶ penalises under-prediction more (chart)

- **DM test** ([Chapter 1](#)): δ_t = loss of model 1 minus loss of model 2

$$DM = \frac{\bar{\delta}}{\sqrt{\hat{\Omega}/T}} \rightarrow N(0, 1)$$

- ▶ $\bar{\delta}$: mean over T forecasts; $\hat{\Omega}$: Newey–West variance $\hat{\gamma}_0 + 2 \sum_{k=1}^L (1 - \frac{k}{L+1}) \hat{\gamma}_k$ of δ_t
- ▶ $DM < -1.96$: model 1 significantly better at 5%

A1: ARFIMA autocorrelations and the mean [Solved]

Task

- ▣ ARFIMA(0, 0.3, 0).
- ▣ 1. Show that $\rho(1) = d/(1 - d)$ and compute $\rho(1)$, $\rho(10)$, $\rho(100)$ with the recursion.
- ▣ 2. Compare $\rho(100)$ with $\frac{\Gamma(1-d)}{\Gamma(d)} 100^{2d-1}$.
- ▣ 3. For $n = 1000$, compute the ratio of the true SD of $\bar{X}$ to the i.i.d. formula $\sqrt{\gamma(0)/n}$ (notebook A1).
- ▣ Report: three numbers; two numbers; one number and one sentence.

Solution

- ▣ 1. $\rho(1) = \rho(0) \cdot \frac{0+d}{1-d}$; $\rho(1) = 0.429$, $\rho(10) = 0.173$, $\rho(100) = 0.069$
- ▣ 2. Asymptotic formula: 0.069, practically equal already at $k = 100$
- ▣ 3. Ratio 7.55: the i.i.d. standard error is 7.55 times too small
- ▣ Interpretation: with $d = 0.3$, 1000 observations carry the information of far fewer independent ones; HAC with a fixed bandwidth cannot repair this

A2: fractional differencing weights [Proposed]

Solution discussed in class

Task

- ▣ $d = 0.4$. Model: A1.
- ▣ 1. Compute $\pi_1, \dots, \pi_5$ of $(1 - L)^{0.4}$.
- ▣ 2. Compare π_{100} with $100^{-d-1}/\Gamma(-d)$.
- ▣ 3. Compute $\sum_{k=0}^{1000} \pi_k$; what is its limit, and what does the gap mean for a truncated filter?
- ▣ Report: five numbers; two numbers; one number and one sentence.

A3: standard errors of GPH and local Whittle [Solved]

Task

- ▣ $n = 5000$, $m = \lfloor n^{0.65} \rfloor$; a local Whittle estimate $\hat{d} = 0.45$.
- ▣ 1. Compute m and the asymptotic SE of GPH and of local Whittle.
- ▣ 2. Give the 95% interval for d and the relative efficiency of LW against GPH.
- ▣ 3. Which m gives a half-width of 0.05?
- ▣ Report: three numbers; an interval and one number; one number.

Solution

- ▣ 1. $m = 253$; GPH $\pi/\sqrt{24m} = 0.0403$; LW $1/(2\sqrt{m}) = 0.0314$
- ▣ 2. $[0.388, 0.512]$; efficiency $(\pi^2/24)/(1/4) = \pi^2/6 = 1.645$
- ▣ 3. $1.96/(2\sqrt{m}) \leq 0.05 \Rightarrow m \geq 385$
- ▣ Interpretation: the interval includes $1/2$, so stationarity is not established; precision is bought with bandwidth, and bandwidth costs bias (A4)

A4: bias and the optimal bandwidth [Proposed]

Solution discussed in class

Task

- ▣ ARFIMA(1, d , 0) with $\phi = 0.5$, $n = 5000$. Model: A3.
- ▣ 1. Compute the bias constant C .
- ▣ 2. For $m = \lfloor n^{0.8} \rfloor$, compute the bias and the SD of local Whittle.
- ▣ 3. Compute m^* , its exponent $\log m^* / \log n$, and the bias and SD at m^* .
- ▣ Report: one number; two numbers; four numbers and one sentence.

A5: fractional Gaussian noise and scaling [Solved]

Task

- ▣ fGn with $H = 0.1, 0.3$ and 0.7 .
- ▣ 1. Compute $\rho(1) = 2^{2H-1} - 1$ and $\rho(2)$ for each H .
- ▣ 2. Compute $m(2, 22)/m(2, 1) = 22^{2H}$: how much more does log volatility move over a month than over a day?
- ▣ 3. Show that $\zeta_q = qH$ for fBm.
- ▣ Report: six numbers; three numbers; one derivation.

Solution

- ▣ 1. $H = 0.1$: $-0.426, -0.026$; $H = 0.3$: $-0.242, -0.049$; $H = 0.7$: $0.320, 0.189$
- ▣ 2. $1.86, 6.39, 75.75$: with $H = 0.1$ the monthly variance of changes is less than twice the daily one
- ▣ 3. $W_{t+\Delta}^H - W_t^H \sim N(0, \Delta^{2H})$: $\mathbb{E}|\cdot|^q = \Delta^{qH} \mathbb{E}|Z|^q$; take logs
- ▣ Interpretation: rough log volatility moves a lot from day to day but drifts little: strong negative correlation of daily changes

A6: the RFSV kernel and the variance correction [Proposed]

Solution discussed in class

Task

- $H = 0.1$, $\nu = 0.3$, horizon $h = 1$; kernel truncated at 500 days. Model: A5.
- 1. Compute the share of the weight on the last day, the last 5 and the last 22 days (notebook A6).
- 2. Compute $c(H)$ and the correction $2c\nu^2 h^{2H}$ for $h = 1, 5, 22$.
- 3. By how much would ignoring the correction understate the variance forecast at $h = 22$?
- Report: three shares; four numbers; one number.

A7: circulant embedding by hand [Solved]

Task

- ▣ Three points of fGn with $H = 0.3$: $\gamma_0, \gamma_1, \gamma_2$.
- ▣ 1. Compute γ_1 and γ_2 .
- ▣ 2. Write the first row of the 4×4 circulant and its eigenvalues $\lambda_k = \sum_j c_j e^{-2\pi ijk/4}$.
- ▣ 3. Can the Davies–Harte method be used?
- ▣ Report: two numbers; a row and four numbers; one sentence.

Solution

- ▣ 1. $\gamma_1 = \frac{1}{2}(2^{0.6} - 2) = -0.2421$,
 $\gamma_2 = \frac{1}{2}(3^{0.6} - 2 \cdot 2^{0.6} + 1) = -0.0491$
- ▣ 2. $c = (\gamma_0, \gamma_1, \gamma_2, \gamma_1)$; $\lambda_0 = \gamma_0 + 2\gamma_1 + \gamma_2 = 0.4666$,
 $\lambda_1 = \lambda_3 = \gamma_0 - \gamma_2 = 1.0491$, $\lambda_2 = \gamma_0 - 2\gamma_1 + \gamma_2 = 1.4352$
- ▣ 3. Yes: all eigenvalues are positive, so the circulant is a valid covariance and its first 3 coordinates have the fGn covariance
- ▣ Interpretation: the embedding doubles the size but turns an $O(n^3)$ factorisation into two FFTs

A8: switching that mimics long memory [Proposed]

Solution discussed in class

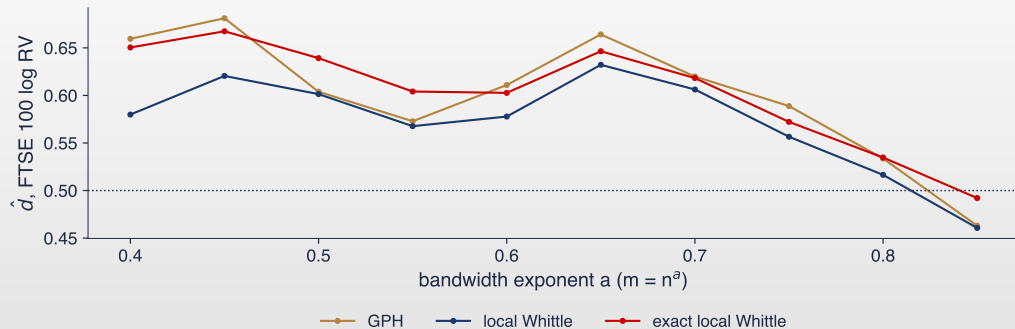
Task

- $X_t = \mu S_t + \varepsilon_t$, $S_t \in \{-1, 1\}$ a Markov chain that stays with probability $q = 0.995$, $\mu = 1$, $\varepsilon_t \sim N(0, 1)$. Model: A1.
- 1. Show that $\rho(k) = \frac{\mu^2}{\mu^2 + 1} (2q - 1)^k$ and compute $\rho(1)$, $\rho(100)$, $\rho(500)$.
- 2. Compute the half-life of the autocorrelation.
- 3. Fit $\log \rho(k)$ on $\log k$ for $k = 5, \dots, 200$ and convert the slope to d .
- Report: three numbers; one number; one number and one sentence.

B1: memory of FTSE 100 realised variance [Solved]

- **Question:** how large is d for the log realised variance of the FTSE 100, and is it robust to the bandwidth and to level shifts?
- **Inputs:** Oxford-Man 5-minute RV of the FTSE 100, $n = 5\,586$ days to 25 February 2022
- **Tasks**
 1. Estimate d of log RV by GPH, local Whittle and exact local Whittle for $m = n^{0.5}, n^{0.6}, n^{0.7}$, with standard errors.
 2. Plot $\hat{d}(m)$ for $m = n^{0.40}, \dots, n^{0.85}$.
 3. Compute the Qu statistic with $m = n^{0.7}$ and $\varepsilon = 0.02$.
 4. Interpretation: is the memory of FTSE 100 volatility genuine, and is it stationary?
- **Report:** a table of nine estimates with SEs, one chart, one statistic, one sentence
- **Notebook:** section B1

B1: solution (1/2) [Solved]



B1: solution (2/2) [Solved]

m	GPH (SE)	LW	ELW	SE (LW, ELW)
74	0.604 (0.075)	0.601	0.639	0.058
177	0.611 (0.048)	0.578	0.603	0.038
419	0.620 (0.031)	0.606	0.618	0.024

- ▣ Qu: $W = 0.33$ against the 10% value 1.09: no evidence of level shifts
- ▣ Interpretation: $\hat{d} \approx 0.6$ at every bandwidth with a flat $\hat{d}(m)$: genuine long memory, at or just beyond the stationarity border; report ELW

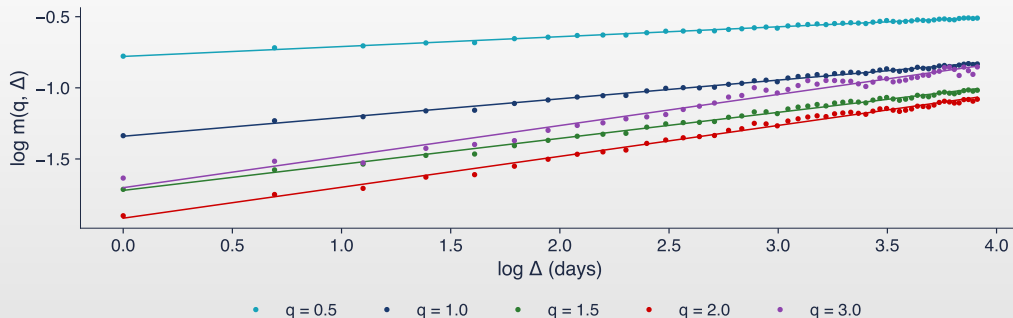
B2: BET and EUR/RON with the Qu test [Proposed]

- **Question:** is the memory of $|r_t|$ on the Bucharest market and for the leu genuine, or does it come from regime changes?
- **Inputs:** BET 2000–2026 and from 2010; EUR/RON (BNR) 2005–2026 and from 2014; model: B1
- **Tasks**
 1. Estimate d of $|r_t|$ by local Whittle with $m = n^{0.65}$ on the full sample and after the cut date.
 2. Compute the Qu statistic ($m = n^{0.7}$, $\varepsilon = 0.02$) on the same samples.
 3. Interpretation: which of the two markets has genuine long memory in volatility?
- **Report:** four estimates with SEs, four statistics, one sentence
- **Notebook:** section B2

B3: roughness of the Euro Stoxx 50 [Solved]

- **Question:** does the Gatheral–Jaisson–Rosenbaum scaling hold for the Euro Stoxx 50, and how rough is its volatility?
- **Inputs:** Oxford-Man 5-minute RV, $n = 5\,635$ days; $\sigma_t = \sqrt{RV_t}$; $q = 0.5, 1, 1.5, 2, 3$; $\Delta = 1, \dots, 50$
- **Tasks**
 1. Compute $m(q, \Delta)$ and the slopes ζ_q ; estimate H by regressing ζ_q on q through the origin.
 2. Estimate H with the measurement-error intercept $m(2, \Delta) = \nu^2 \Delta^{2H} + 2s^2$.
 3. Repeat the first estimate on the two halves of the sample.
 4. Interpretation: is ζ_q linear in q , and what does a departure mean?
- **Report:** five slopes, three values of H , ν , one sentence
- **Notebook:** section B3

B3: solution [Solved]



□ $\zeta_q = 0.069, 0.132, 0.183, 0.217, 0.218$; $\hat{H} = 0.093$, $\hat{\nu} = 0.38$; with the error intercept 0.109; halves 0.091 and 0.095

□ Interpretation: linear up to $q = 2$, flat between $q = 2$ and 3: large moments are driven by a few extreme days (heavy tails of the increments), so H from $q \leq 2$ is the safer reading; roughness is stable over time

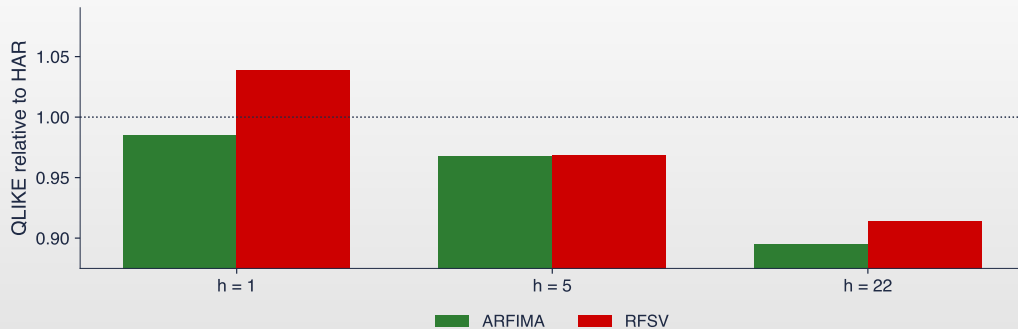
B4: Bitcoin roughness by sampling interval [Proposed]

- **Question:** does the estimate of H for Bitcoin depend on the sampling interval of the realised variance?
- **Inputs:** Binance RV of Bitcoin from 1-, 5-, 15-, 30- and 60-minute returns, 2018–2026; model: B3
- **Tasks**
 1. Estimate H for each RV by the moment regression and with the error intercept.
 2. Estimate d of log RV by local Whittle with $m = n^{0.65}$.
 3. Interpretation: which direction does a coarser grid push H , and why?
- **Report:** ten values of H , five values of d , one sentence
- **Notebook:** section B4

B5: forecasting FTSE 100 realised variance [Solved]

- **Question:** does a rough or a long-memory model forecast FTSE 100 realised variance better than HAR?
- **Inputs:** rolling window of 1000 days, re-estimation every 20 days; out of sample 30 December 2003 – 26 January 2022 ($T = 4\,565$); horizons 1, 5, 22 days
- **Tasks**
 1. Forecast RV_{t+h} with HAR, ARFIMA(0, d , 0) and RFSV, with the lognormal corrections.
 2. Compute the average QLIKE of each model and the ratio to HAR.
 3. Compute the DM statistics against HAR.
 4. Interpretation: which model would you use at each horizon?
- **Report:** nine losses, six ratios, six DM statistics, one sentence
- **Notebook:** section B5

B5: solution (1/2) [Solved]



B5: solution (2/2) [Solved]

h	QLIKE HAR	ARFIMA / HAR	DM	RFSV / HAR	DM
1	0.230	0.985	-0.98	1.038	1.49
5	0.324	0.968	-1.94	0.968	-1.33
22	0.561	0.895	-2.12	0.914	-1.93

- Interpretation: HAR at one day (RFSV is worse); at 22 days both ARFIMA ($p = 0.034$) and RFSV ($p = 0.053$) beat HAR by about a tenth; median rolling $\hat{H} = 0.11$

B6: forecasting CAC 40 realised variance [Proposed]

- **Question:** does the FTSE 100 ranking carry over to the CAC 40?
- **Inputs:** the same design, out of sample 22 December 2003 – 26 January 2022 ($T = 4\,631$); model: B5
- **Tasks**
 1. Repeat the three forecasts and the QLIKE ratios.
 2. Compute the DM statistics against HAR.
 3. Interpretation: is there a single best model across the two indices?
- **Report:** six ratios, six DM statistics, one sentence
- **Notebook:** section B6

C1: co-memory of DAX and CAC 40 volatility [Proposed]

- **Question:** do the volatilities of two neighbouring markets share one fractional trend?
- **Inputs:** log 5-minute RV of the DAX and the CAC 40 (Oxford-Man), $n = 5\,600$ common days; FCVAR with $k = 0$; model: B1
- **Tasks**
 1. Write a five-line pre-registration: hypothesis, model, grid of (d, b) , tests, what would change your mind.
 2. Estimate d of each series and of the spread $\beta'X_t$ by local Whittle, then the FCVAR by profile likelihood with trace tests.
 3. Interpretation: what do the data support, and what would you add to the model?
- **Report:** a five-line plan, five estimates, two tests, a project plan
- **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant to summarise long memory and rough volatility. The answer:
- ▣ (a) An ARFIMA with $d = 0.3$ has autocorrelations that decay exponentially, only more slowly than an AR(1).
- ▣ (b) The standard error of local Whittle is $\pi/\sqrt{24m}$.
- ▣ (c) A larger bandwidth m is always better, because the variance falls.
- ▣ (d) If the Qu test rejects, the series has structural breaks.
- ▣ (e) $H = 0.1$ means that volatility has short memory, because $H < 0.5$ is antipersistence.
- ▣ (f) FIGARCH with $0 < d < 1$ is covariance stationary with a finite unconditional variance.
- ▣ (g) Exact local Whittle remains consistent for $d = 1.2$, whereas local Whittle does not.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
 - ▶ 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- ▣ Long memory makes i.i.d. and HAC standard errors far too small
- ▣ Local Whittle is the default; exact local Whittle when d may exceed $1/2$; always vary m
- ▣ Test against level shifts before claiming long memory
- ▣ Log volatility is rough ($H \approx 0.1$) and persistent; the proxy affects H
- ▣ Forecast gains over HAR are small and horizon-dependent

After the seminar

- ▣ Lecture 10 develops each topic of today: spectral theory, asymptotics of the estimators, the Qu test, FCVAR, FIGARCH, rough volatility and its forecasts
- ▣ Try the [Proposed] tasks in the notebook
- ▣ C1 can grow into a team project: co-memory of Central and Eastern European volatilities
- ▣ Reading: [Robinson \(1995b\)](#); [Shimotsu and Phillips \(2005\)](#); [Qu \(2011\)](#); [Gatheral, Jaisson and Rosenbaum \(2018\)](#)
- ▣ **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

References (1/2)

- Baillie, R. T., Bollerslev, T., & Mikkelsen, H. O. (1996). Fractionally integrated generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 74(1), 3–30. doi:10.1016/S0304-4076(95)01749-6
- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196. doi:10.1093/jjfinec/nbp001
- Davies, R. B., & Harte, D. S. (1987). Tests for Hurst effect. *Biometrika*, 74(1), 95–101. doi:10.1093/biomet/74.1.95
- Diebold, F. X., & Inoue, A. (2001). Long memory and regime switching. *Journal of Econometrics*, 105(1), 131–159. doi:10.1016/S0304-4076(01)00073-2
- Gatheral, J., Jaisson, T., & Rosenbaum, M. (2018). Volatility is rough. *Quantitative Finance*, 18(6), 933–949. doi:10.1080/14697688.2017.1393551
- Geweke, J., & Porter-Hudak, S. (1983). The estimation and application of long memory time series models. *Journal of Time Series Analysis*, 4(4), 221–238. doi:10.1111/j.1467-9892.1983.tb00371.x
- Heber, G., Lunde, A., Shephard, N., & Sheppard, K. K. (2009). Oxford-Man Institute's realized library, version 0.3. Oxford-Man Institute, University of Oxford. web.archive.org
- Hosking, J. R. M. (1981). Fractional differencing. *Biometrika*, 68(1), 165–176. doi:10.1093/biomet/68.1.165

References (2/2)

- ▣ Johansen, S., & Nielsen, M. Ø. (2012). Likelihood inference for a fractionally cointegrated vector autoregressive model. *Econometrica*, 80(6), 2667–2732. doi:10.3982/ECTA9299
- ▣ Mandelbrot, B. B., & Van Ness, J. W. (1968). Fractional Brownian motions, fractional noises and applications. *SIAM Review*, 10(4), 422–437. doi:10.1137/1010093
- ▣ Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256. doi:10.1016/j.jeconom.2010.03.034
- ▣ Qu, Z. (2011). A test against spurious long memory. *Journal of Business & Economic Statistics*, 29(3), 423–438. doi:10.1198/jbes.2010.09153
- ▣ Robinson, P. M. (1995b). Gaussian semiparametric estimation of long range dependence. *The Annals of Statistics*, 23(5), 1630–1661. doi:10.1214/aos/1176324317
- ▣ Shimotsu, K. (2010). Exact local Whittle estimation of fractional integration with unknown mean and time trend. *Econometric Theory*, 26(2), 501–540. doi:10.1017/S0266466609100075
- ▣ Shimotsu, K., & Phillips, P. C. B. (2005). Exact local Whittle estimation of fractional integration. *The Annals of Statistics*, 33(4), 1890–1933. doi:10.1214/009053605000000309